

Implementation of U-Net Architecture for Cryptocurrency Time Series Forecasting

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1 Introduction

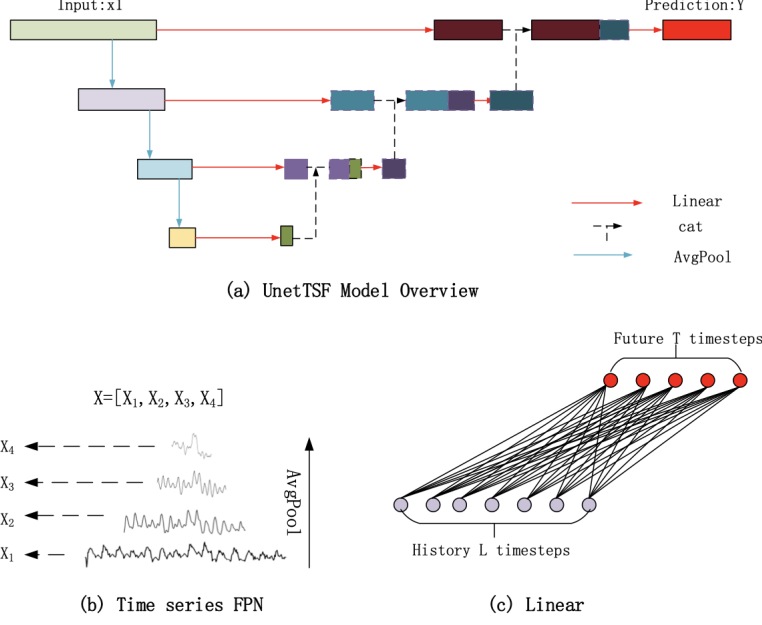
This research focuses on the reimplementation of a popular U-Net architecture first designed for image segmentation purposes in the medical industry in [2]. The paper follows the pipeline and the design proposed in [1], where the original architecture is modified for time series prediction purposes. In [1] it is experimentally shown that the U-Net architecture was superior to other commonly used methods (see [1] for details) with improved results among various datasets, including ETTh1, ETTh2, ETTm1.

My personal interest lies in assessing the ability of the proposed architecture to accurately predict highly volatile time series in the context of crypto data, that seemingly, has no inherent structure in terms of its seasonality or trend, and if it does try to generate alpha using the results.

First to make sure the architecture is consistent, I first test the architecture on the same datasets as ETTh1, ETTh2, ETTm1, and compare the results. The cryptocurrency data consist of hourly close prices quoted in USDT (note that predicting the price is similar in complexity is predicting derived returns for future alpha generation, that is experimenting on price suffices).

The predictive tasks follows the work in [1] directly. Given the last L hours of an asset's price, the model forecasts the next T hours. The model builds a hierarchy of temporal resolutions via pooling, makes a separate prediction at each scale, and then fuses these multi-scale predictions into a single forecast over the desired horizon. The result is a compact model that is still able to capture multi-scale temporal structure.

Importantly, I do not assume stationarity: the model is retrained on a rolling window of recent data at each decision time, so that both the forecasts adapt to evolving market structure. This setup, although intentionally minimal, demonstrates the full path from time series modelling to backtesting.



2 Model Architecture

As mentioned in the introduction, the model architecture is a modified, even simplified, version of the original U-Net proposed in [2], and follows the architecture described in [1]. Historically, the key element in time series analysis, was decomposing the data into: Trend, Cycle, Seasonality and Random components. This architecture, attempts to incorporate a similar in spirit idea, to construct, via average pooling applied on the 1D time series, the feature pyramid network. This allows for the model to extract features in accordance to difference scales in data, akin to how time series was historically analysed, in methods like ARIMA. For the purposes of this paper, in particular to predict both volatility and returns, I have modified the original network to input along two time series instead of one. For training and testing, the network takes as input tensors of shape (B, C, L) , where B is the batch size, $C = 1$ corresponds to the time series channel (I have also constructed architecture that is robust to predict along multiple channels), and L is the length of the lookback window(see Figure ??). The encoder builds a "feature" pyramid by repeatedly applying one-dimensional average pooling on subsequent time series. In other words, starting from the input $x^{(0)}$, the original series, the encoder produces

$$x^{(0)}, x^{(1)}, \dots, x^{(N-1)},$$

with $N = 4$ in our case, and where at each stage, except the last, the next level is obtained by

$$x^{(i+1)} = \text{AvgPool1d}(x^{(i)}),$$

with a **kernel size of three and a stride of two**. Each pooling operation reduces the effective temporal resolution and decreases the length of the time series, analogous to how spatial pooling in image models produces coarser feature maps.

At each level i in the pyramid, the model constructs a linear predictions of length T , denoted $y^{(i)}$. To get the final prediction, the model incorporates the following fusion mechanism. Starting from $y^{(N-1)}$ (dimension (B, C, T)), we construct the fused predictions $\hat{y}^{(i)}$ from bottom up, with $y^{(N-1)} = \hat{y}^{(N-1)}$. To obtain $\hat{y}^{(i-1)}$ from $y^{(i)}$, the model concatenate $\hat{y}^{(i-1)}$ with $y^{(i)}$ resulting in output of dimension $(B, 2C, T)$. This tensor is then interpreted as $B \cdot T$

vectors in $\mathbb{R}^{2C}$, each corresponding to a single time step and batch element, and passed through a fusion linear layer that maps $\mathbb{R}^{2C}$ back to $\mathbb{R}^C$. Reshaping and permuting dimensions back restores the tensor to shape (B, C, T) . This process effectively learns, for each forecast horizon and time step, how to combine the “coarse” and “fine” predictions into a single prediction. The final output of the network is the fused forecast at the shallowest level.

3 Results

We are gonna start with benchmark data sets ETTh1, ETTh2, ETTm1 to test the model’s consistency for further experimentation. The following details were obtained.

Table 1: Results for Different Datasets with $T = 96$

Dataset	Train Loss	Validation MSE		Win Count	Total Samples	Win Rate (%)
		U-Net	Martingale			
ETTh1	0.1487	0.069024	0.076673	1193	2167	55.053
ETTh2	0.1658	0.155084	0.307308	1602	2182	73.419
ETTM1	0.0847	0.039512	0.040377	5423	10021	54.116
Total/Aggregate:					14370	57.189

This confirms that the U-Net architecture showed very similar results to those presented in [1], and outperformed the martingale prediction ($\hat{x}_t = x_{t-1}$) consistently among all three datasets. This allows to be able to make clear judgment for the crypto dataset.

Having validated the U-Net-based architecture on the benchmark datasets from [1], where it consistently outperformed the martingale baseline $\hat{x}_t = x_{t-1}$, we now turn to the more challenging setting of hourly cryptocurrency prices. Tables 2–5 report, for ten major crypto pairs and a range of lookback windows L and horizons T , the training loss, the out-of-sample MSE on the validation set, and the fraction of validation windows where U-Net beats the martingale predictor.

It is very easy to see that the U-Net has failed to outperform the martingale predictor in a prevailing majority of cases, and indeed, for short horizons ($T = 5$ and $T = 10$) the win rates cluster around 30%–45% for most assets. Only in a few cases (e.g. SOLUSDT or LINKUSDT at certain (L, T) combinations) does the win rate approach or slightly exceed 50%, and even then the improvement over the baseline is marginal in absolute terms. For longer horizons ($T = 24$ and $T = 96$) the picture remains mixed: some assets (for example ADAUSDT or LINKUSDT at $T = 96$) show win rates just above 50%, but others (BNBUSDT, TRXUSDT and, to a lesser extent, BTCUSDT) exhibit significantly worse MSE than the martingale, with win rates often well below 30%.

The dependence on the lookback window L is also instructive. Shorter frames such as $L = 24$ or $L = 96$ generally perform as well as the longest window $L = 336$. As L increases, the U-Net MSE often deteriorates relative to the martingale baseline (e.g. BTCUSDT and TRXUSDT at $L = 336$ for all T). This suggests that, for these series, very long histories do not translate into additional predictive power for the architecture; instead, they may introduce

already irrelevant data into the analysis, making analysis harder, especially given the aggressive downsampling in the FPN-style encoder.

Overall, the results are negative from a pure forecasting perspective: unlike the benchmark datasets used in [1], the hourly crypto prices do not demonstrate any gain from replacing the martingale with the U-Net model in terms of squared-error prediction of future prices, not to mentioned more complex models, like ARIMA.

The contrast between the strong performance reported in [1] on standard forecasting benchmarks and the weak or inconsistent gains observed here on crypto prices can be explained by several structural differences in the data.

First, many of the benchmark datasets in [1] (electricity load, traffic, exchange rates, ETT, etc.) possess strong and relatively stable patterns: daily and weekly seasonality, smooth trends, and regimes that evolve slowly over time. A multi-scale architecture such as Time-Series U-Net is well suited to exploit such regularities: average pooling can compress local patterns, and multi-level fusion can capture interactions across time scales. Hhourly crypto price series are much more volatile. If the underlying process is approximately a random walk with fat tails, then the martingale predictor $\hat{x}_{t+k} = x_t$ is close to optimal in MSE for short horizons. In that regime, even a sophisticated architecture might not do much better than that, because there is very little predictable structure to exploit. The small but systematic gaps between the U-Net and martingale MSE (in favour of the martingale) are consistent with this picture. Using MSE as the sole loss function allows large price jumps dominate the optimization. Occasional large errors on volatility spikes can significantly inflate MSE, even if the model captures some softer patterns such as mild mean reversion or intraday structure. Perhaps identifying the outliers in the data first, prior to using the model, like using ensemble methods, might improve the predictive power.

Finally, the current evaluation focuses on predicting prices directly. For an almost-martingale series, predicting the level is a much easier task for the naive baseline than for a model that tries to infer structure in the *increments*. From a financial perspective, what matters for trading is predictive power on returns or other transformed quantities (spreads, volatility, directions), not on the level of the process; in that sense, pure price-level MSE is an exceptionally tough metric to improve upon.

4 Conclusion and Future Work

The experiments carried out in this project show that the Time-Series U-Net architecture, while capable of matching the behaviour reported in [1] on structured benchmark datasets, does not deliver a predictive advantage over a martingale baseline when applied to hourly cryptocurrency prices. Across a wide range of lookback lengths L and prediction horizons T , the U-Net’s validation MSE on price forecasts remains very close to that of the naive $\hat{x}_t = x_{t-1}$ predictor, in fact falling short of it majority of cases.

From a modelling point of view, this negative result is itself informative. It suggests that, under the information set considered here (historical prices only, at an hourly frequency), the crypto assets studied behave close to a random walk at the time scales of interest. In such a regime, sophisticated sequence models have limited room to extract additional predictive structure beyond what is already captured by the trivial baseline. The sensitivity of performance to L and T , and the mixed results across assets, further indicate that any exploitable predictability, if present, is weak,

unstable, and heavily overshadowed by noise and regime changes.

There are several natural directions for future work that could strengthen, refine, or challenge these conclusions. First, predicting returns and volatility rather than prices. A more financially meaningful setup would operate directly on (log-)returns and possibly model volatility as a separate target channel. This is closer to the architecture discussed in [1] .

Secondly , incorporating volume, order-book proxies, realized volatility measures, or macro/market-structure signals might provide the model with more stable predictors than price alone, but are not available for me in "at home " conditions. Regime-aware training (for example via regime-switching labels, clustering, or explicit volatility regimes) could mitigate the impact of abrupt structural breaks on the learned filters. In summary, the current experiments suggest that, for hourly crypto prices, the Time-Series U-Net does not deliver a strong improvement over a simple martingale baseline. This is less a failure of the architecture per se than a reflection of the statistical properties of the underlying data: high volatility, heavy tails, and limited exploitable structure at the horizons studied.

References

- [1] Li Chu, Xiao Bingjia, and Yuan Qiping. Unettsf: A better performance linear complexity time series prediction model, 2024.
- [2] Olaf Ronneberger, Philipp Fischer, and Thomas Brox. U-net: Convolutional networks for biomedical image segmentation, 2015.

Table 2: Results for $T = 5$ with different L values

Asset	L	Train Loss	MSE		Win Count	Total Samples	Win Rate (%)
			U-Net	Martingale			
ADAUSDT	24	0.0070	0.002068	0.002048	541	1285	42.101
	96	0.0073	0.002345	0.002110	493	1213	40.643
	92	0.0072	0.002233	0.002107	527	1217	43.303
	336	0.0077	0.002773	0.002359	361	973	37.102
BNBUSDT	24	0.0184	0.012659	0.012325	594	1285	46.226
	96	0.0186	0.014845	0.012793	464	1213	38.252
	92	0.0186	0.013936	0.012774	538	1217	44.207
	336	0.0193	0.027579	0.014522	258	973	26.516
BTCUSDT	24	0.0044	0.002723	0.002404	479	1285	37.276
	96	0.0045	0.002883	0.002387	364	1213	30.008
	92	0.0046	0.002724	0.002385	444	1217	36.483
	336	0.0049	0.005155	0.002224	206	973	21.172
DOGEUSDT	24	0.0038	0.001282	0.001190	532	1285	41.401
	96	0.0040	0.001388	0.001232	476	1213	39.242
	92	0.0040	0.001392	0.001231	456	1217	37.469
	336	0.0045	0.001773	0.001410	246	973	25.283
DOTUSDT	24	0.0052	0.001381	0.000912	384	1285	29.883
	96	0.0054	0.001092	0.000934	421	1213	34.707
	92	0.0055	0.001118	0.000933	418	1217	34.347
	336	0.0059	0.001561	0.001001	301	973	30.935
ETHUSDT	24	0.0035	0.002934	0.002777	555	1285	43.191
	96	0.0036	0.003819	0.002797	456	1213	37.593
	92	0.0036	0.003087	0.002799	477	1217	39.195
	336	0.0041	0.003718	0.002968	337	973	34.635
LINKUSDT	24	0.0068	0.002275	0.002164	545	1285	42.412
	96	0.0072	0.002346	0.002221	527	1213	43.446
	92	0.0070	0.002749	0.002218	436	1217	35.826
	336	0.0080	0.002895	0.002389	273	973	28.058
SOLUSDT	24	0.0065	0.002754	0.002674	604	1285	47.004
	96	0.0069	0.002936	0.002639	449	1213	37.016
	92	0.0069	0.002878	0.002636	486	1217	39.934
	336	0.0072	0.003096	0.002730	370	973	38.027
TRXUSDT	24	0.0086	0.002787	0.002231	308	1285	23.969
	96	0.0089	0.002345	0.002142	521	1213	42.951
	92	0.0089	0.005418	0.002140	190	1217	15.612
	336	0.0092	0.003887	0.002329	182	973	18.705
XRPUSDT	24	0.0033	0.002484	0.002440	571	1285	44.436
	96	0.0034	0.002606	0.002519	526	1213	43.364
	92	0.0035	0.002689	0.002513	476	1217	39.113
	336	0.0040	0.005542	0.002892	224	973	23.022

Table 3: Results for $T = 10$ with different L values

Asset	L	Train Loss	MSE		Win Count	Total Samples	Win Rate (%)
			U-Net	Martingale			
ADAUSDT	24	0.0127	0.003730	0.003674	574	1280	44.844
	96	0.0130	0.004127	0.003792	474	1208	39.238
	92	0.0130	0.004028	0.003783	492	1212	40.594
	336	0.0136	0.004601	0.004243	412	968	42.562
BNBUSDT	24	0.0326	0.025929	0.021944	494	1280	38.594
	96	0.0328	0.027710	0.022776	407	1208	33.692
	92	0.0330	0.028591	0.022746	545	1212	44.967
	336	0.0338	0.029958	0.025605	356	968	36.777
BTCUSDT	24	0.0078	0.005871	0.004469	454	1280	35.469
	96	0.0080	0.004888	0.004426	420	1208	34.768
	92	0.0080	0.005119	0.004421	358	1212	29.538
	336	0.0084	0.010004	0.004169	198	968	20.455
DOGEUSDT	24	0.0067	0.002235	0.002148	588	1280	45.938
	96	0.0068	0.002529	0.002233	398	1208	32.947
	92	0.0068	0.002533	0.002229	480	1212	39.604
	336	0.0073	0.003097	0.002565	286	968	29.545
DOTUSDT	24	0.0092	0.001741	0.001663	510	1280	39.844
	96	0.0093	0.002098	0.001699	373	1208	30.877
	92	0.0094	0.001954	0.001695	409	1212	33.746
	336	0.0101	0.002512	0.001843	306	968	31.612
ETHUSDT	24	0.0062	0.005485	0.005114	537	1280	41.953
	96	0.0067	0.006139	0.005196	467	1208	38.659
	92	0.0064	0.005673	0.005191	474	1212	39.109
	336	0.0076	0.006177	0.005484	376	968	38.843
LINKUSDT	24	0.0119	0.004078	0.004042	596	1280	46.562
	96	0.0122	0.004447	0.004146	539	1208	44.619
	92	0.0123	0.004497	0.004136	490	1212	40.429
	336	0.0133	0.005225	0.004464	320	968	33.058
SOLUSDT	24	0.0116	0.004963	0.004816	626	1280	48.906
	96	0.0119	0.005163	0.004770	519	1208	42.964
	92	0.0118	0.004995	0.004758	554	1212	45.710
	336	0.0125	0.005492	0.004954	425	968	43.905
TRXUSDT	24	0.0174	0.005042	0.003982	370	1280	28.906
	96	0.0178	0.007903	0.003835	207	1208	17.136
	92	0.0177	0.005179	0.003828	300	1212	24.752
	336	0.0185	0.005492	0.004181	214	968	22.107
XRPUSDT	24	0.0058	0.004703	0.004549	528	1280	41.250
	96	0.0059	0.005093	0.004673	481	1208	39.818
	92	0.0059	0.004918	0.004660	522	1212	43.069
	336	0.0062	0.005689	0.005344	417	968	43.079

Table 4: Results for $T = 24$ with different L values

Asset	L	Train Loss	MSE		Win Count	Total Samples	Win Rate (%)
			U-Net	Martingale			
ADAUSDT	24	0.0282	0.008615	0.008521	559	1266	44.155
	96	0.0290	0.009061	0.008767	504	1194	42.211
	92	0.0290	0.009027	0.008744	513	1198	42.821
	336	0.0288	0.010772	0.009914	446	954	46.751
BNBUSDT	24	0.0696	0.058889	0.050574	487	1266	38.468
	96	0.0704	0.062564	0.052833	438	1194	36.683
	92	0.0706	0.060567	0.052703	435	1198	36.311
	336	0.0712	0.072090	0.060500	328	954	34.382
BTCUSDT	24	0.0166	0.011850	0.010326	437	1266	34.518
	96	0.0169	0.011224	0.010142	390	1194	32.663
	92	0.0168	0.010998	0.010122	426	1198	35.559
	336	0.0169	0.011882	0.010106	339	954	35.535
DOGEUSDT	24	0.0143	0.005012	0.004856	539	1266	42.575
	96	0.0145	0.005426	0.005062	516	1194	43.216
	92	0.0142	0.005328	0.005049	458	1198	38.230
	336	0.0148	0.006941	0.005884	325	954	34.067
DOTUSDT	24	0.0202	0.004058	0.003822	488	1266	38.547
	96	0.0202	0.005778	0.003868	369	1194	30.905
	92	0.0198	0.004508	0.003859	513	1198	42.821
	336	0.0205	0.006273	0.004202	343	954	35.954
ETHUSDT	24	0.0142	0.013916	0.012185	521	1266	41.153
	96	0.0145	0.013930	0.012434	472	1194	39.531
	92	0.0142	0.013376	0.012405	479	1198	39.983
	336	0.0152	0.014613	0.013227	385	954	40.356
LINKUSDT	24	0.0262	0.010542	0.009402	474	1266	37.441
	96	0.0268	0.010129	0.009575	493	1194	41.290
	92	0.0265	0.009863	0.009548	521	1198	43.489
	336	0.0267	0.011047	0.010254	423	954	44.340
SOLUSDT	24	0.0254	0.011475	0.011340	604	1266	47.709
	96	0.0259	0.011329	0.011137	550	1194	46.064
	92	0.0255	0.011135	0.011106	562	1198	46.912
	336	0.0266	0.012338	0.011692	449	954	47.065
TRXUSDT	24	0.0442	0.012773	0.008542	305	1266	24.092
	96	0.0448	0.009594	0.008507	339	1194	28.392
	92	0.0455	0.011295	0.008491	227	1198	18.948
	336	0.0450	0.019850	0.009411	146	954	15.304
XRPUSDT	24	0.0126	0.011401	0.011083	519	1266	40.995
	96	0.0130	0.011818	0.011356	526	1194	44.054
	92	0.0129	0.011748	0.011320	508	1198	42.404
	336	0.0131	0.013349	0.013018	430	954	45.073

Table 5: Results for $T = 96$ with different L values

Asset	L	Train Loss	MSE		Win Count	Total Samples	Win Rate (%)
			U-Net	Martingale			
ADAUSDT	24	0.0914	0.034620	0.036094	620	1194	51.926
	96	0.0930	0.035474	0.037251	582	1122	51.872
	92	0.0927	0.035670	0.037140	582	1126	51.687
	336	0.0866	0.047701	0.043488	453	882	51.361
BNBUSDT	24	0.1995	0.246435	0.197247	357	1194	29.899
	96	0.2020	0.261440	0.207927	308	1122	27.451
	92	0.2018	0.241261	0.207218	364	1126	32.327
	336	0.2033	0.418660	0.244128	166	882	18.821
BTCUSDT	24	0.0542	0.043354	0.037492	401	1194	33.585
	96	0.0551	0.039197	0.038441	483	1122	43.048
	92	0.0549	0.038108	0.038321	494	1126	43.872
	336	0.0538	0.047497	0.037774	210	882	23.810
DOGEUSDT	24	0.0590	0.023342	0.024237	580	1194	48.576
	96	0.0598	0.024446	0.025535	560	1122	49.911
	92	0.0595	0.024377	0.025450	541	1126	48.046
	336	0.0589	0.032204	0.030629	388	882	43.991
DOTUSDT	24	0.0794	0.023544	0.015334	369	1194	30.905
	96	0.0807	0.018273	0.015268	452	1122	40.285
	92	0.0810	0.020809	0.015252	386	1126	34.281
	336	0.0731	0.030908	0.016778	299	882	33.900
ETHUSDT	24	0.0512	0.061723	0.061219	582	1194	48.744
	96	0.0520	0.067715	0.064257	473	1122	42.157
	92	0.0520	0.068820	0.064049	488	1126	43.339
	336	0.0528	0.075991	0.071614	323	882	36.621
LINKUSDT	24	0.0940	0.034627	0.037469	666	1194	55.779
	96	0.0958	0.036111	0.039097	660	1122	58.824
	92	0.0953	0.037594	0.038969	586	1126	52.043
	336	0.0890	0.040949	0.042424	554	882	62.812
SOLUSDT	24	0.0877	0.042872	0.044895	678	1194	56.784
	96	0.0893	0.042158	0.045561	681	1122	60.695
	92	0.0891	0.042852	0.045433	655	1126	58.171
	336	0.0895	0.048191	0.048040	468	882	53.061
TRXUSDT	24	0.1404	0.067591	0.031507	186	1194	15.578
	96	0.1409	0.100494	0.032863	132	1122	11.765
	92	0.1406	0.056730	0.032753	201	1126	17.851
	336	0.1383	0.050776	0.039110	171	882	19.388
XRPUSDT	24	0.0447	0.053381	0.050285	521	1194	43.635
	96	0.0450	0.051628	0.052865	645	1122	57.487
	92	0.0450	0.054761	0.052682	537	1126	47.691
	336	0.0456	0.068761	0.063202	347	882	39.342